

Pandas Functions for AI/ML Students

Creating DataFrames & Series

`pd.Series()`, `pd.DataFrame()`, `pd.Index()`

Reading Data

`read_csv()`, `read_excel()`, `read_json()`, `read_parquet()`, `read_sql()`, `read_html()`, `read_pickle()`

Saving Data

`to_csv()`, `to_excel()`, `to_json()`, `to_parquet()`, `to_sql()`, `to_pickle()`

Data Inspection

`head()`, `tail()`, `sample()`, `shape`, `size`, `columns`, `index`, `dtypes`, `info()`, `describe()`, `memory_usage()`, `value_counts()`, `nunique()`, `unique()`

Selecting Data

`loc[]`, `iloc[]`, `at[]`, `iat[]`, `filter()`, `query()`

Sorting

`sort_values()`, `sort_index()`, `nlargest()`, `nsmallest()`, `rank()`

Filtering

`query()`, `isin()`, `between()`, boolean indexing

Missing Values

`isnull()`, `isna()`, `notnull()`, `dropna()`, `fillna()`, `replace()`, `interpolate()`

Duplicates

`duplicated()`, `drop_duplicates()`

Rename

`rename()`, `set_axis()`

Columns

`insert()`, `assign()`, `drop()`, `pop()`

Math & Stats

`sum()`, `mean()`, `median()`, `mode()`, `std()`, `var()`, `min()`, `max()`, `count()`, `prod()`, `quantile()`, `abs()`, `round()`, `corr()`, `cov()`, `corrwith()`, `skew()`, `kurt()`

String

`str.lower()`, `str.upper()`, `str.title()`, `str.capitalize()`, `str.strip()`, `str.replace()`, `str.contains()`, `str.startswith()`, `str.endswith()`, `str.split()`, `str.join()`, `str.extract()`, `str.len()`, `str.count()`

Datetime

`pd.to_datetime()`, `dt.year`, `dt.month`, `dt.day`, `dt.hour`, `dt.minute`, `dt.second`, `dt.day_name()`, `dt.month_name()`, `dt.weekday`, `dt.date`, `dt.time`

Apply Family

`apply()`, `map()`, `transform()`, `pipe()`

GroupBy

`groupby()`, `agg()`, `transform()`, `filter()`, `size()`, `count()`

Combine

`merge()`, `join()`, `concat()`, `combine_first()`

Pivot

`pivot()`, `pivot_table()`, `crosstab()`, `melt()`, `stack()`, `unstack()`

Window

`rolling()`, `expanding()`, `ewm()`

Feature Engineering

`cut()`, `qcut()`, `get_dummies()`, `factorize()`, `astype()`, `clip()`

Index

`reset_index()`, `set_index()`, `reindex()`, `sort_index()`

Boolean

`any()`, `all()`, `where()`, `mask()`

Sampling

`sample()`

Explode

`explode()`

Reshaping

`transpose()`, `T`, `swaplevel()`

Time Series

`resample()`, `shift()`, `diff()`, `pct_change()`, `asfreq()`

Utilities

`copy()`, `compare()`, `equals()`, `infer_objects()`, `convert_dtypes()`

Top 30

`read_csv()`, `head()`, `info()`, `describe()`, `loc`, `iloc`, `isnull()`, `fillna()`, `dropna()`, `drop_duplicates()`,
`value_counts()`, `unique()`, `groupby()`, `merge()`, `concat()`, `pivot_table()`, `sort_values()`, `query()`, `apply()`,
`map()`, `get_dummies()`, `factorize()`, `corr()`, `astype()`, `to_datetime()`, `sample()`, `reset_index()`,
`set_index()`, `to_csv()`